

Operations Research

Sensitivity and Reoptimization

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Course outline

- **Linear Optimization**

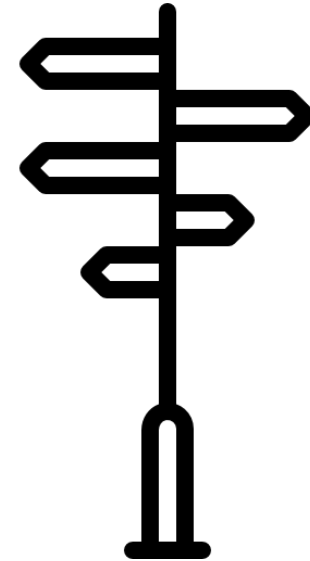
- Modeling with linear programming and graphical solution
- Solving linear programs, convexity
- Simplex algorithm, 2-phase Simplex
- Simplex algorithm in matrix notation
- **Sensitivity analysis, reoptimization**
- Duality theory, zero-sum games

- **Integer Linear Optimization**

- Modeling integer programming problems (IPs)
- Solving IPs: branch & bound, cutting planes
- Advanced solution methods: column generation, approximation algorithms
- Fast solvable integer problems: total unimodularity, matroids
- Network flow problems
- Traveling salesperson problem

- **Nonlinear Optimization**

- Optimization of multidimensional functions
- Convex optimization



Today you will learn...

- how to analyze the **effect of minor changes in the parameters of an LP**.
- how to determine the **value of your resources**.
- how to **reoptimize** LPs after adding new variables.
- how to perform **sensitivity analysis with Gurobi**.

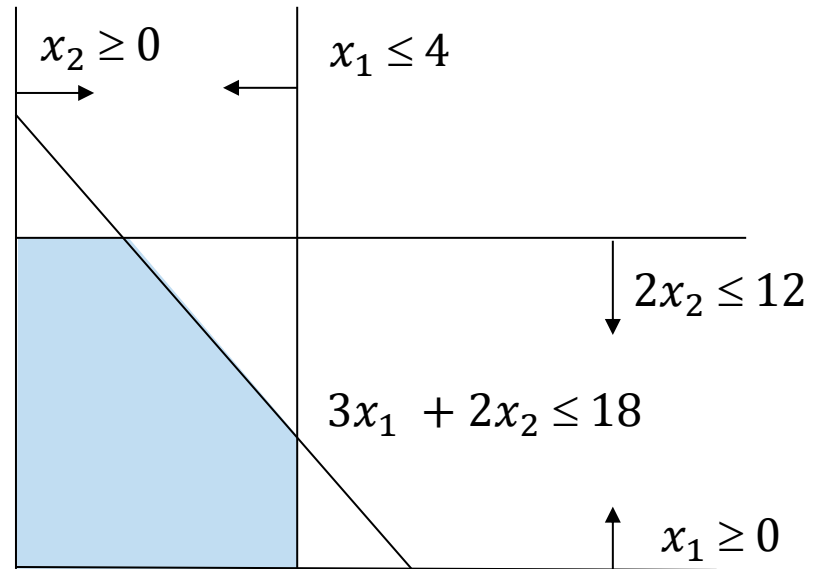


Example - production

	Product		
	1 (Profit: 5)	2 (Profit: 3)	
Machine	Production hours per truckload		Available hours/week
1	1	0	4
2	0	2	12
3	3	2	18

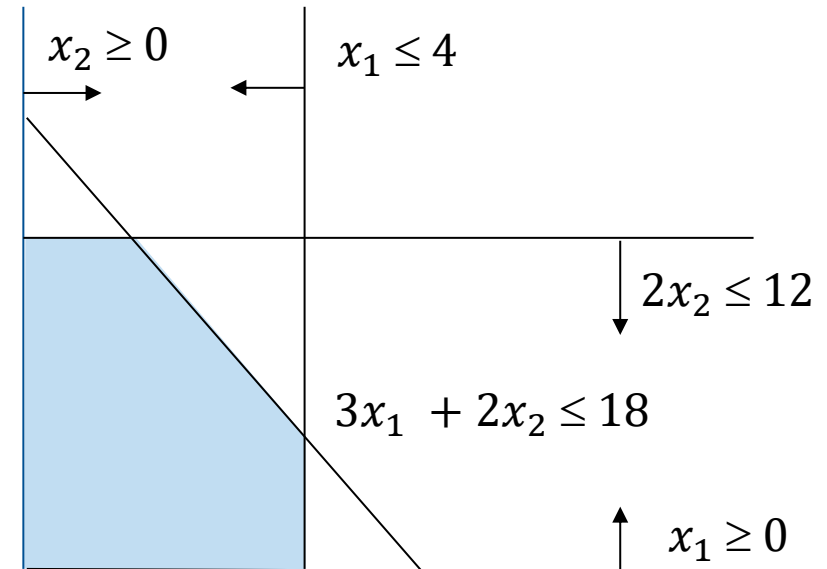
Linear program

$$\begin{array}{llll} \max & 5x_1 & + & 3x_2 \\ \text{s.t.} & x_1 & & \leq 4 \\ & & 2x_2 & \leq 12 \\ & 3x_1 & + & 2x_2 \leq 18 \\ & x_1, x_2 & \geq & 0 \end{array}$$



Linear program in standard form

$$\begin{array}{llllll}
 \max & 5x_1 & + & 3x_2 & & \\
 \text{s.t.} & x_1 & & & + & x_3 & = & 4 \\
 & & 2x_2 & & + & x_4 & = & 12 \\
 & 3x_1 & + & 2x_2 & & & + & x_5 & = & 18 \\
 & & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0
 \end{array}$$



The LP solution is $(4, 3, 0, 6, 0)$ with basic variables $\{x_1, x_2, x_4\}$ and an objective value **29**.

Solution via (revised) Simplex

$$\begin{aligned}
 \max \quad & 5x_1 + 3x_2 \\
 \text{s.t.} \quad & x_1 + x_3 = 4 \\
 & 2x_2 + x_4 = 12 \\
 & 3x_1 + 2x_2 + x_5 = 18 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0
 \end{aligned}$$

Basic variables	Coefficients						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1			$1/2$		$3/2$	29
x_1		1		1		0	4
x_2			1	$-3/2$		$1/2$	3
x_4				3	1	-1	6

Formulas from the previous class

$$\begin{aligned}
 \max \quad & 5x_1 + 3x_2 \\
 \text{s.t.} \quad & x_1 + x_3 = 4 \\
 & 2x_2 + x_4 = 12 \\
 & 3x_1 + 2x_2 + x_5 = 18 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0
 \end{aligned}$$

Basic variables $\{x_1, x_2, x_4\}$

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 3 & 1 & -1 \end{pmatrix}$$

Basic variable	Coefficients							Solution
	Z	x_{B_1}	...	x_{B_m}	x_{N_1}	...	$x_{N_{n-m}}$	
Z	1	0		0	$\mathbf{c}_N'^T = -(\mathbf{c}_N - \mathbf{N}'^T \mathbf{c}_B)^T$			$\mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{b}$
x_{B_1}		1			$\mathbf{N}' = \mathbf{B}^{-1} \mathbf{N}$			$\mathbf{b}' = \mathbf{B}^{-1} \mathbf{b}$
...			...					
x_{B_m}				1				

Sensitivity analysis

How do changes in the data affect the feasibility and optimality of the basis?

- What if machine 1 can run longer?

Change in the vector $\mathbf{b}$

- What happens if the profit of the first product is not 5 € but only 3 €?

Change in the vector $\mathbf{c}$

It holds:

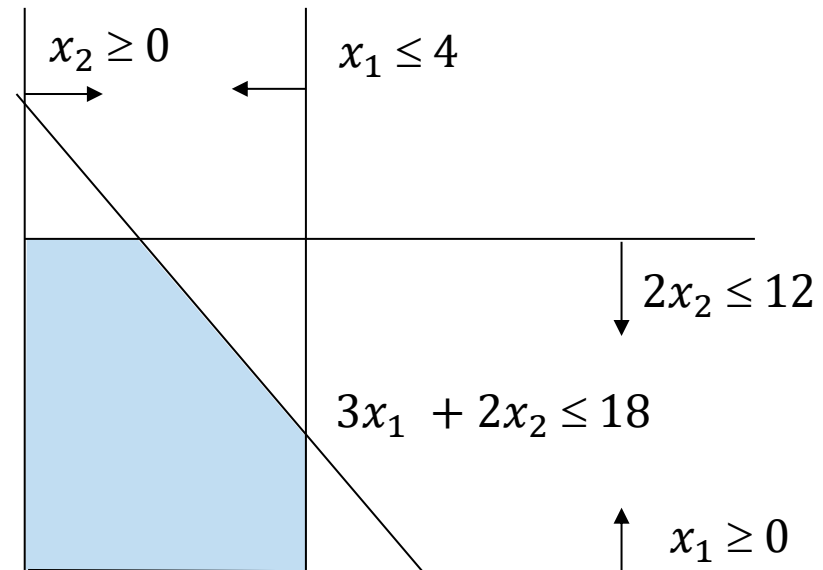
The basis is feasible if $\mathbf{B}^{-1}\mathbf{b} \geq \mathbf{0}$

The basis is optimal if $-(\mathbf{c}_N - \mathbf{N}'^T \mathbf{c}_B) \geq \mathbf{0}$

Change on the right hand side

Example: If machine 1 can run for more hours during the week?

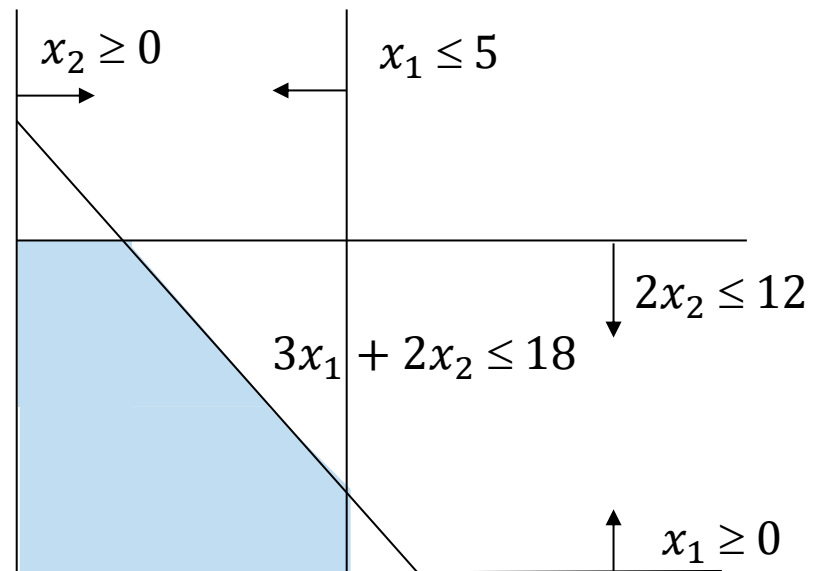
$$\begin{array}{llll}
 \max & 5x_1 & + & 3x_2 \\
 \text{s.t.} & x_1 & & \leq 4 \\
 & & 2x_2 & \leq 12 \\
 & 3x_1 & + & 2x_2 \leq 18 \\
 & & x_1, x_2 & \geq 0
 \end{array}$$



Change on the right hand side

Example: If machine 1 can run for more hours during the week?

$$\begin{array}{llll} \max & 5x_1 & + & 3x_2 \\ \text{s.t.} & x_1 & & \leq 5 \\ & & 2x_2 & \leq 12 \\ & 3x_1 & + & 2x_2 \leq 18 \\ & & x_1, x_2 & \geq 0 \end{array}$$



What is changed in the result?

$$\begin{aligned}
 \max \quad & 5x_1 + 3x_2 \\
 \text{s.t.} \quad & x_1 + x_3 = 5 \\
 & 2x_2 + x_4 = 12 \\
 & 3x_1 + 2x_2 + x_5 = 18 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0
 \end{aligned}$$

Basic variable	Coefficients							Solution
	Z	x_{B_1}	...	x_{B_m}	x_{N_1}	...	$x_{N_{n-m}}$	
Z	1	0		0	$c'_N = -(c_N - N'^T c_B)^T$			$c_B^T B^{-1} b$
x_{B_1}		1			$N' = B^{-1} N$			$b' = B^{-1} b$
...			...					
x_{B_m}				1				

Solution via (revised) Simplex

$$\begin{aligned}
 \max \quad & 5x_1 + 3x_2 \\
 \text{s.t.} \quad & x_1 + x_3 = 5 \\
 & 2x_2 + x_4 = 12 \\
 & 3x_1 + 2x_2 + x_5 = 18 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0
 \end{aligned}$$

Basic variables $\{x_1, x_2, x_4\}$

$$c_B^T B^{-1} = (5 \ 3 \ 0) * \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 3 & 1 & -1 \end{pmatrix} = \left(\frac{1}{2}, 0, \frac{3}{2}\right)$$

$$c_B^T B^{-1} b = \left(\frac{1}{2} \ 0 \ \frac{3}{2}\right) * \begin{pmatrix} 5 \\ 12 \\ 18 \end{pmatrix} = 29.5$$

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1			$1/2$		$3/2$	29.5
x_1		1		1		0	5
x_2			1	$-3/2$		$1/2$	1.5
x_4				3	1	-1	9

Shadow prices

The entries under the slack variables contained in the result row of each simplex tableau are called opportunity costs (shadow prices).

Basic variable	Coefficient						Solution
	z	x_1	x_2	x_3	x_4	x_5	
z	1			$1/2$		$3/2$	29.5
x_1		1		1		0	5
x_2			1	$-3/2$		$1/2$	1.5
x_4				3	1	-1	9

$$c'_N = -(c_N - (B^{-1}N)^T c_B)^T = -c_N^T + c_B^T B^{-1} N \text{ and } c_B^T B^{-1} = \left(\frac{1}{2}, 0, \frac{3}{2}\right)$$

$$c'_N = -(0 \ 0) + \left(\frac{1}{2}, 0, \frac{3}{2}\right) \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} = \left(\frac{1}{2}, \frac{3}{2}\right)$$

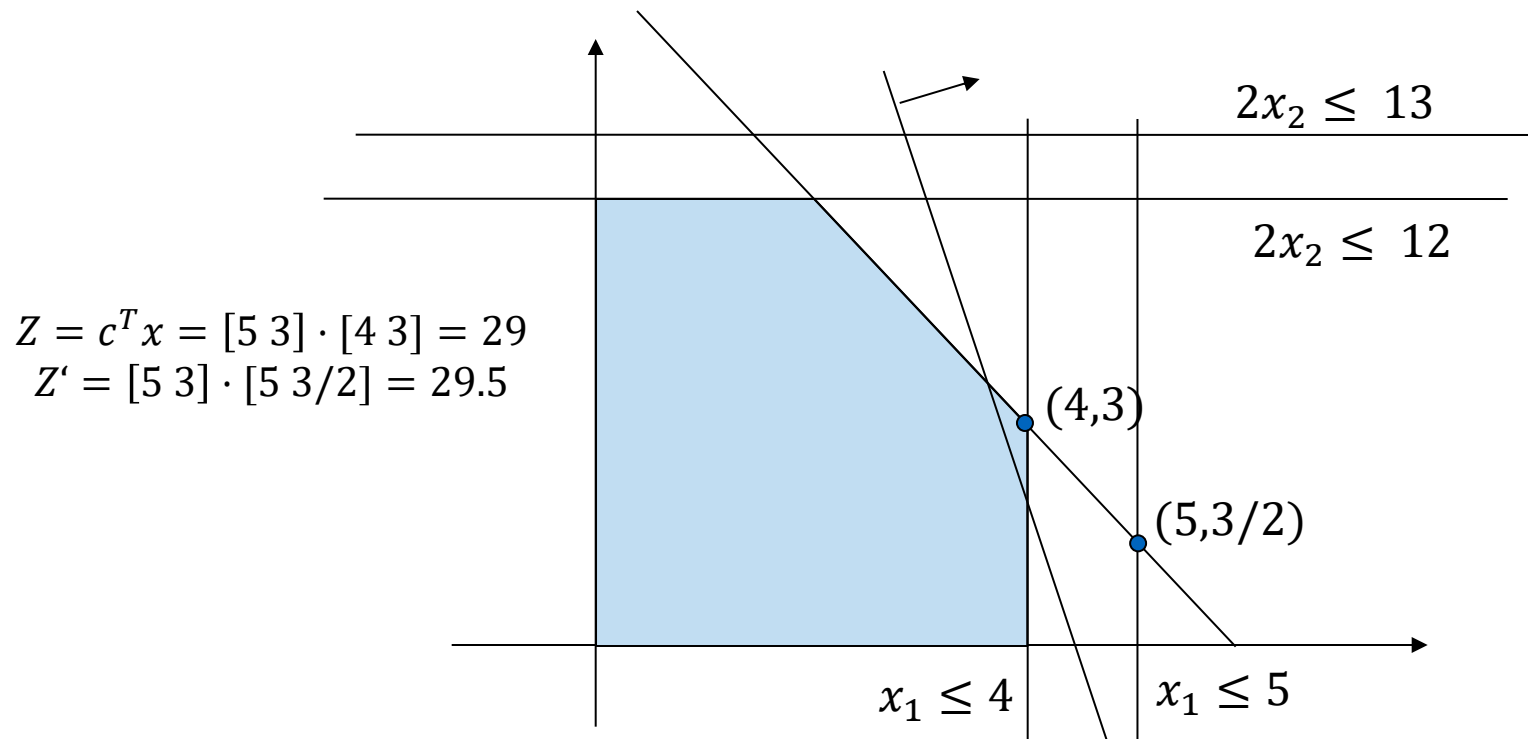
Shadow prices

$$\begin{array}{llllll}
 \max & 5x_1 & + & 3x_2 & & \\
 \text{s.t.} & x_1 & & & + & x_3 & = & 4 \\
 & & 2x_2 & & + & x_4 & = & 12 \\
 & 3x_1 & + & 2x_2 & & & + & x_5 & = & 18 \\
 & & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0
 \end{array}$$

- x_3 is the slack variable in the capacity constraint for machine 1.
- If the capacity is increased from 4 to 5, we obtain the point $(5, 3/2)$ as the optimal solution with an objective function value of 29.5.
- The profit gain for an additional unit of time on $M1$ corresponds to the value $\frac{1}{2}$ that can be read in the simplex tableau.

Shadow prices

Resource constraints with shadow prices are called binding constraints.



Shadow prices

Many LPs are used to allocate scarce resources to activities.

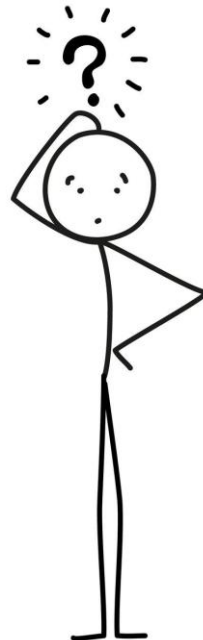
The shadow price of a resource is the marginal value of an additional unit of this resource to the firm.

The shadow price of a resource with non-binding outcomes is 0.

Binding constraints correspond to scarce resources and therefore have positive shadow prices/opportunity costs.

This information is important in deciding which resources are best to invest in.

You are doing shift scheduling at an airport (as in one of the earlier modeling examples) and have a constraint per hour that specifies how many employees must be present over multiple shifts. How should the shadow price of this constraint be interpreted?



Roster

Time period	Time frame					Minimum number of required workers
	Shift					
	1	2	3	4	5	
6 am to 8 am	x					48
8 am to 10 am	x	x				79
10 am to 12 am	x	x				65
12 am to 2 pm	x	x	x			87
2 pm to 4 pm		x	x			64
4 pm to 6 pm			x	x		73
6 pm to 8 pm			x	x		82
8 pm to 10 pm				x		43
10 pm to 12 pm				x	x	52
12 pm to 6 am					x	15

Check feasibility for changes in b

For which values on the right hand side does the current basis remain feasible?

It must hold that $B^{-1}b \geq 0$!

$$B^{-1}b \geq 0 \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 3 & 0 & 1 \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ 3 & 1 & -1 \end{pmatrix} \begin{pmatrix} 4 + \delta \\ 12 \\ 18 \end{pmatrix} \geq 0$$

$$\begin{pmatrix} 4 + \delta \\ 3 - 3\delta/2 \\ 6 + 3\delta \end{pmatrix} \geq 0 \Rightarrow -2 \leq \delta \leq 2$$

The optimal basis is feasible as long as the number of hours on machine 1 is between 2 and 6.

→ i.e. the shadow price of 0.5 €/hour corresponding to machine 1 is valid as long as the number of hours is between **2** and **6**.

Shadow prices for the given b

$$B^{-1}b \geq 0 \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 3 & 0 & \frac{1}{2} \\ -\frac{2}{2} & 0 & \frac{2}{2} \\ 3 & 1 & -1 \end{pmatrix} \begin{pmatrix} 4 + \delta \\ 12 \\ 18 \end{pmatrix} \geq 0 \quad \begin{pmatrix} 4 + \delta \\ 3 - 3\delta/2 \\ 6 + 3\delta \end{pmatrix} \geq 0 \Rightarrow -2 \leq \delta \leq 2$$

$$N' = B^{-1}N = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 3 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{3}{2} & \frac{1}{2} \\ 3 & -1 \end{pmatrix}$$

$$c'_N = -(c_N - N'^T c_B)^T = -\left((0 \ 0) - \begin{pmatrix} 1 & -\frac{3}{2} & 3 \\ 0 & \frac{1}{2} & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \end{pmatrix} = (x_3 \ x_5)$$

Shadow price of machine 1 for < 2 hours

$$\begin{array}{llllll}
 \max & 5x_1 & + & 3x_2 & & \\
 \text{s.t.} & x_1 & & & + & x_3 & = & \mathbf{1} \\
 & & & 2x_2 & & + & x_4 & = & 12 \\
 & 3x_1 & + & 2x_2 & & & + & x_5 & = & 18 \\
 & & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0
 \end{array}$$

Old basis variables $\{x_1, x_2, x_4\}$

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 3 & 1 & -1 \end{pmatrix}$$

$$B^{-1}b = (1, \frac{15}{2}, -3) \text{ infeasible}$$

Now another basis is optimal. The shadow prices for the basis x_1, x_2, x_5 are

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ -3 & -1 & 1 \end{pmatrix}, N' = B^{-1} N = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \\ -3 & -1 \end{pmatrix}$$

$$c'_N = -(c_N - N'^T c_B)^T = (5, 3/2) \Rightarrow \{x_3, x_4\}$$

The shadow price for machine 1 is now 5 €, while the shadow price for machine 2 is 3/2.

Shadow prices in Gurobi

Just like variables, attributes of constraints can be queried after optimization:

```
model.addConstr(... , name='c1')  
c1 = model.getConstrByName('c1')
```

Query shadow prices:

```
c1.Pi
```

Querying the bounds of the shadow prices:

```
c1.SARHSLow  
c1.SARHSUp
```

Gurobi example

```
model = gp.Model('ProductionPlan')

# Create variables
x1 = model.addVar(lb=0, ub=GRB.INFINITY, vtype=GRB.CONTINUOUS, name='Product 1')
x2 = model.addVar(lb=0, ub=GRB.INFINITY, vtype=GRB.CONTINUOUS, name='Product 2')

# Add constraint:  $x_1 \leq 4$ 
model.addConstr(x1 <= 4)

# Add constraint:  $2x_2 \leq 12$ 
model.addConstr(2 * x2 <= 12)

# Add constraint:  $3x_1 + 2x_2 \leq 18$ 
model.addConstr(3 * x1 + 2 * x2 <= 18)

# Set objective
model.setObjective(5 * x1 + 3 * x2, GRB.MAXIMIZE)

# Optimize model
model.optimize()
```


Gurobi example

```
print(f'Objective = {model.getObjective().getValue()}')
print(f'{x1.VarName} = {x1.X}')
print(f'{x2.VarName} = {x2.X}')

# Get shadow prices
m1 = model.getConstrByName('Machine 1')
m2 = model.getConstrByName('Machine 2')
m3 = model.getConstrByName('Machine 3')

print(f'Shadow price for constraint 1: {m1.Pi} between {m1.SARHSLow} and {m1.SARHSUp}')
print(f'Shadow price for constraint 2: {m2.Pi} between {m2.SARHSLow} and {m2.SARHSUp}')
print(f'Shadow price for constraint 3: {m3.Pi} between {m3.SARHSLow} and {m3.SARHSUp}')
```

Objective = 29.0

Product 1 = 4.0

Product 2 = 3.0

Shadow price for constraint 1: 0.5 between 2.0 and 6.0

Shadow price for constraint 2: 0.0 between 6.0 and inf

Shadow price for constraint 3: 1.5 between 12.0 and 24.0

Parametric programming in Gurobi

```
rhs_val = 0
m1.rhs = rhs_val
model.optimize()
shadow = m1.Pi

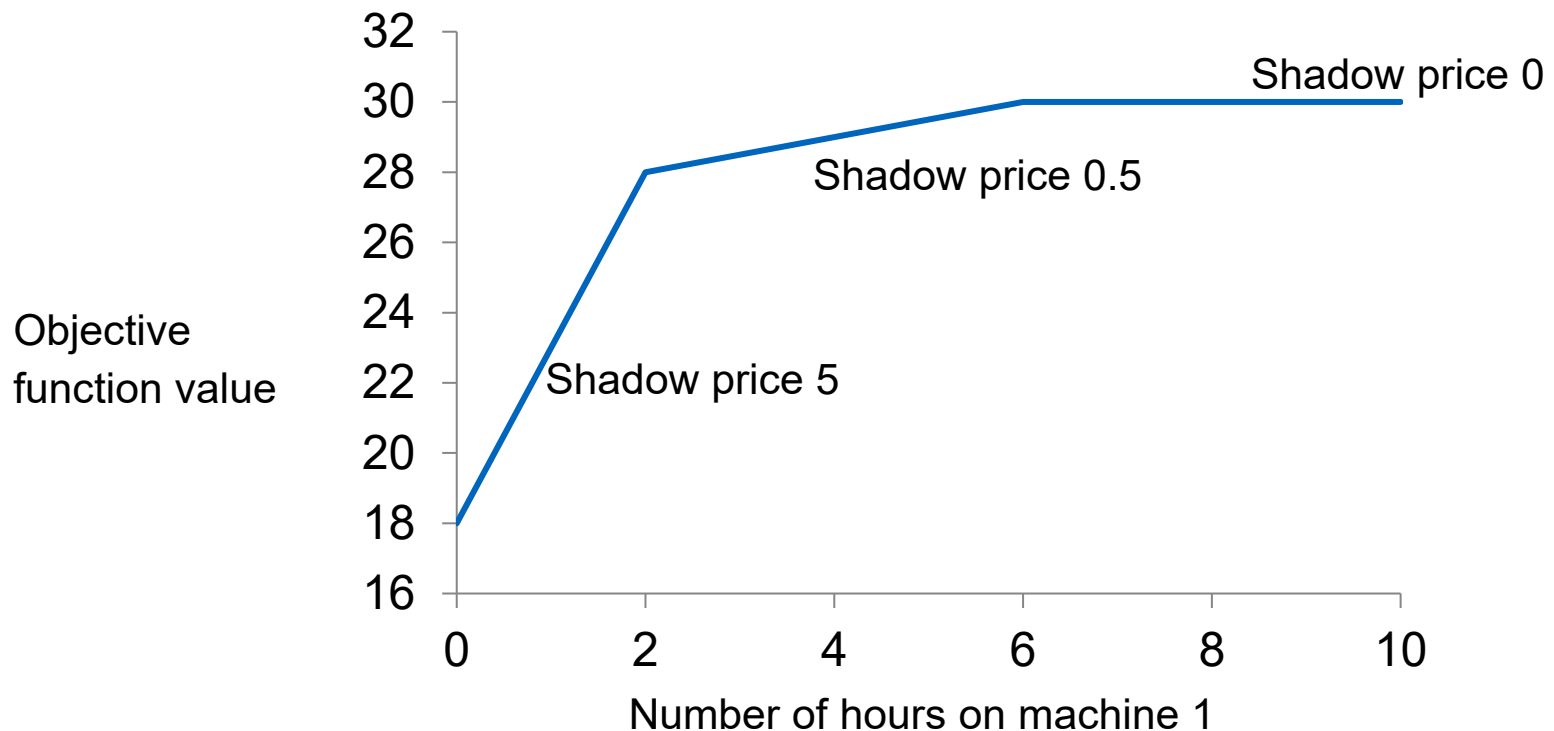
while shadow > 0:
    print(f'Shadow price for constraint 1: {m1.Pi} between {m1.SARHSLow} and {m1.SARHSUp}')
    rhs_val = m1.SARHSUp
    m1.rhs = rhs_val + 0.01
    model.update()
    model.optimize()
    shadow = m1.Pi

print(f'Shadow price for constraint 1: {m1.Pi} between {m1.SARHSLow} and {m1.SARHSUp}')
```

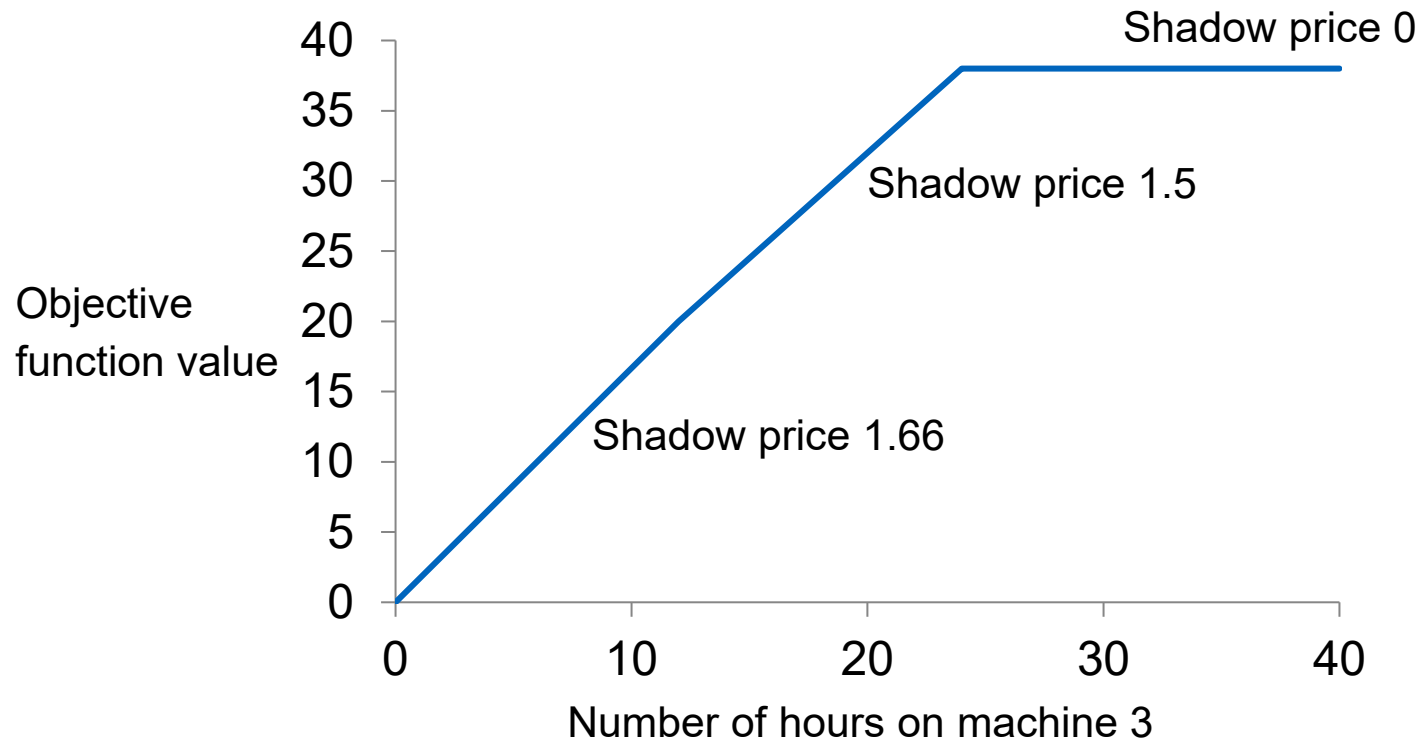
- Shadow price of constraint 1: 5.0 for RHS between 0.0 and 2.0
- Shadow price of constraint 1: 0.5 for RHS between 2.0 and 6.0
- Shadow price of constraint 1: 0.0 for RHS between 6.0 and 1.0E100

Parametric programming

- What is the objective function value depending on the available hours on machine 1?
- This can be computed by iterative analysis analogously to the previous slides.



Machine 3



Change in two constraints

Suppose you can reallocate hours between machines 1 and 3.
What would be the best distribution of hours?

$$B^{-1}b \geq 0 \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ -3/2 & 0 & 1/2 \\ 3 & 1 & -1 \end{pmatrix} \begin{pmatrix} 4 + \delta \\ 12 \\ 18 - \delta \end{pmatrix} \geq 0$$

$$\begin{pmatrix} 4 + \delta \\ 3 - 2\delta \\ 6 + 4\delta \end{pmatrix} \geq 0 \Rightarrow -3/2 \leq \delta \leq 3/2$$

Change in the objective function value in this range: $-3/2$ is optimal

$$c_B^T B^{-1}b = (B^{-1}b)^T c_B = \begin{pmatrix} 4 + \delta \\ 3 - 2\delta \\ 6 + 4\delta \end{pmatrix}^T \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} = 29 - \delta$$

Change in two constraints

$$\begin{matrix} x_1 \\ x_2 \\ x_4 \end{matrix} \begin{pmatrix} 4 + \delta \\ 3 - 2\delta \\ 6 + 4\delta \end{pmatrix} \geq 0 \Rightarrow -3/2 \leq \delta \leq 3/2 \quad \begin{matrix} x_3 & x_5 \\ N' = \begin{pmatrix} 1 & 0 \\ -3/2 & 1/2 \\ 3 & -1 \end{pmatrix} \begin{matrix} x_1 \\ x_2 \\ x_4 \end{matrix} \\ N' = B^{-1}N \end{matrix}$$

For $\delta < -3/2$, x_4 is negative: basis change of x_4 and x_5

$$B^{-1}b = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ -3 & -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 + \delta \\ 12 \\ 18 - \delta \end{pmatrix} \geq 0 \quad \begin{pmatrix} 4 + \delta \\ 6 \\ -6 - 4\delta \end{pmatrix} \geq 0 \Rightarrow -4 \leq \delta \leq -3/2$$

Change in the objective value in this range:

$$c_B^T B^{-1}b = (B^{-1}b)^T c_B = \begin{pmatrix} 4 + \delta \\ 6 \\ -6 - 4\delta \end{pmatrix}^T \cdot \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix} = 38 + 5\delta$$

Since only 4 hours are available on machine 1, the problem becomes infeasible for $\delta < -4$.

Change in two constraints

$$\begin{matrix} x_1 \\ x_2 \\ x_4 \end{matrix} \begin{pmatrix} 4 + \delta \\ 3 - 2\delta \\ 6 + 4\delta \end{pmatrix} \geq 0 \Rightarrow -3/2 \leq \delta \leq 3/2 \quad \begin{matrix} x_3 & x_5 \\ N' = \begin{pmatrix} 1 & 0 \\ -3/2 & 1/2 \\ 3 & -1 \end{pmatrix} \begin{matrix} x_1 \\ x_2 \\ x_4 \end{matrix} \\ N' = B^{-1}N \end{matrix}$$

For $\delta > 3/2$, x_2 is negative: basis change of x_2 and x_3

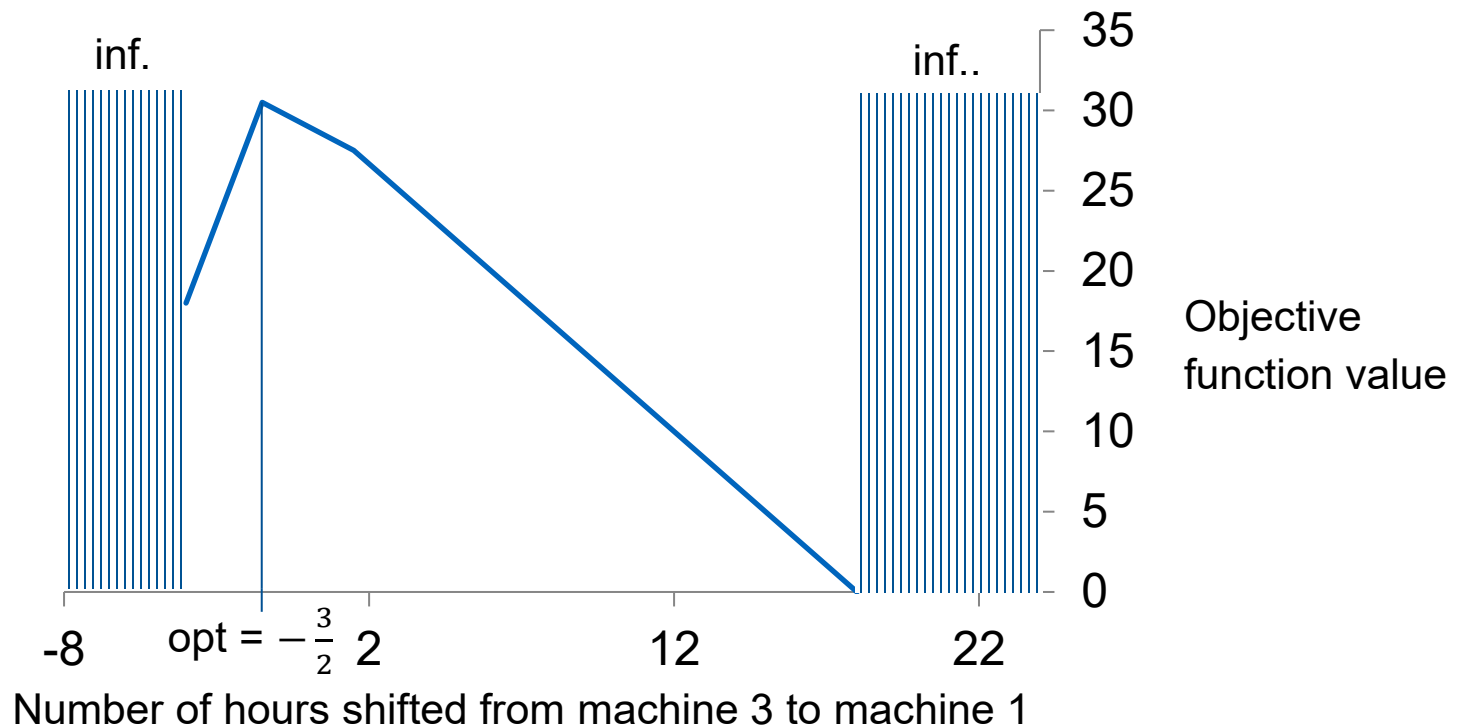
$$\begin{pmatrix} 0 & 0 & 1/3 \\ 1 & 0 & -1/3 \\ 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 + \delta \\ 12 \\ 18 - \delta \end{pmatrix} \geq 0 \quad \begin{pmatrix} 6 - 1/3 \delta \\ -2 + 4/3 \delta \\ 12 \end{pmatrix} \geq 0 \Rightarrow \frac{3}{2} \leq \delta \leq 18$$

Change in the objective value in this range:

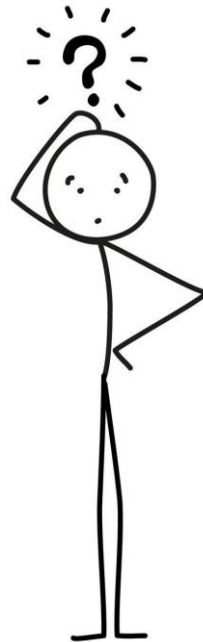
$$c_B^T B^{-1} b = (B^{-1} b)^T c_B = \begin{pmatrix} 6 - 1/3 \delta \\ -2 + 4/3 \delta \\ 12 \end{pmatrix}^T \cdot \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} = 30 - 5/3 \delta$$

Since only 18 hours are available on machine 3, the problem becomes infeasible for $\delta > 18$.

Parametric programming



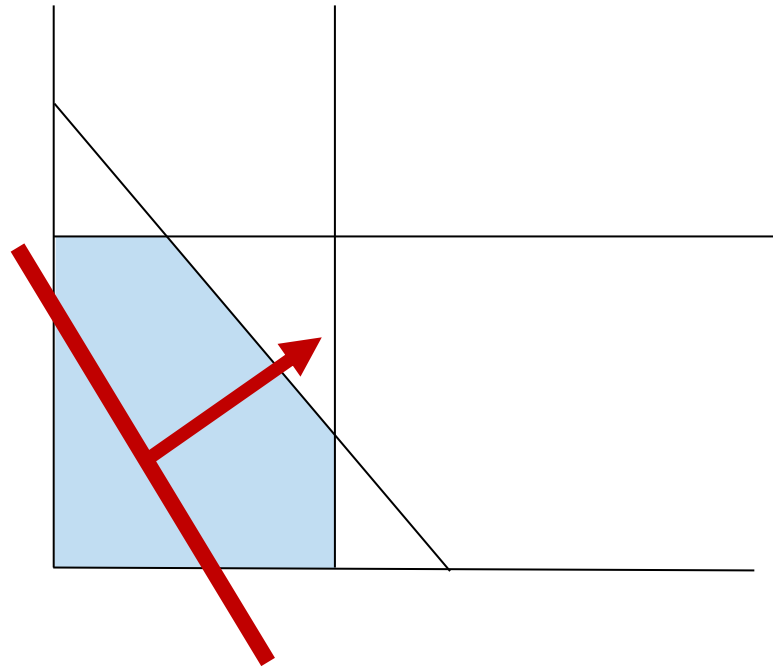
What are shadow prices and when are they valid?



Change in the objective function

Example: The profit of the first product drops from 5 € to 3 €.

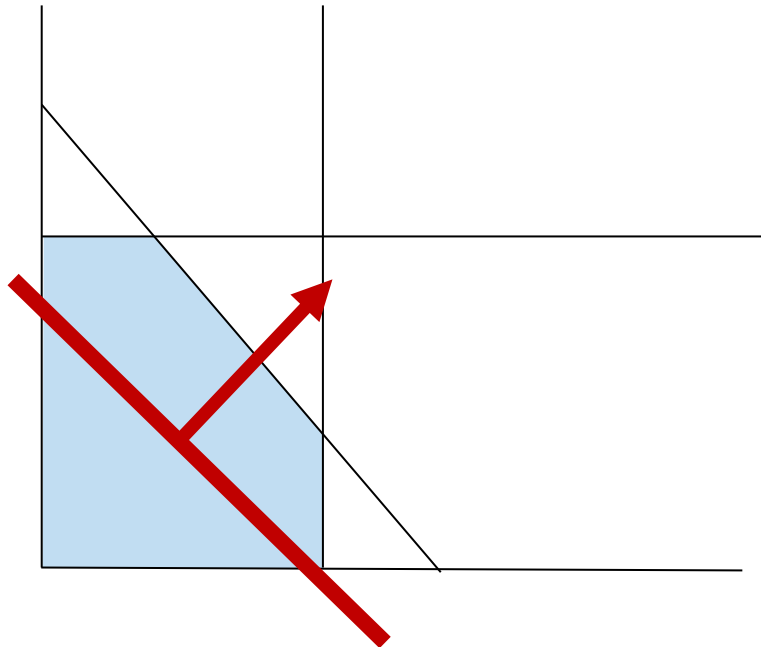
$$\begin{array}{llll} \max & 5x_1 & + & 3x_2 \\ \text{s.t.} & x_1 & & \leq 4 \\ & & 2x_2 & \leq 12 \\ & 3x_1 & + & 2x_2 \leq 18 \\ & x_1, x_2 & \geq & \end{array}$$



Change in the objective function

Example: The profit of the first product drops from 5 € to 3 €.

$$\begin{array}{llll} \max & 3x_1 & + & 3x_2 \\ \text{s.t.} & x_1 & & \leq 4 \\ & & 2x_2 & \leq 12 \\ & 3x_1 & + & 2x_2 \leq 18 \\ & x_1, x_2 & \geq & \end{array}$$



What changes in the result?

Basic variables	Coefficients:							Solution
	Z	x_{B_1}	...	x_{B_m}	x_{N_1}	...	$x_{N_{n-m}}$	
Z	1	0		0	$\mathbf{c}_N'^T = -(c_N - N'^T c_B)^T$			$\mathbf{c}_B^T B^{-1} b$
x_{B_1}		1						$b' = B^{-1} b$
...			...					
x_{B_m}				1				
					$N' = B^{-1} N$			

Change in the objective function

$$x_B = (x_1 \ x_2 \ x_4)^T, \quad \mathbf{c}_B = (\mathbf{3 \ 3 \ 0}), \quad x_N = (x_3 \ x_5)^T, \quad c_N = (0 \ 0)$$

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{pmatrix}, N = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Is the current basis $\{x_1, x_2, x_4\}$ still optimal?

Compute $-(c_N - N'^T c_B)$

Example

Compute $c'_N = -(c_N - N'^T c_B)$

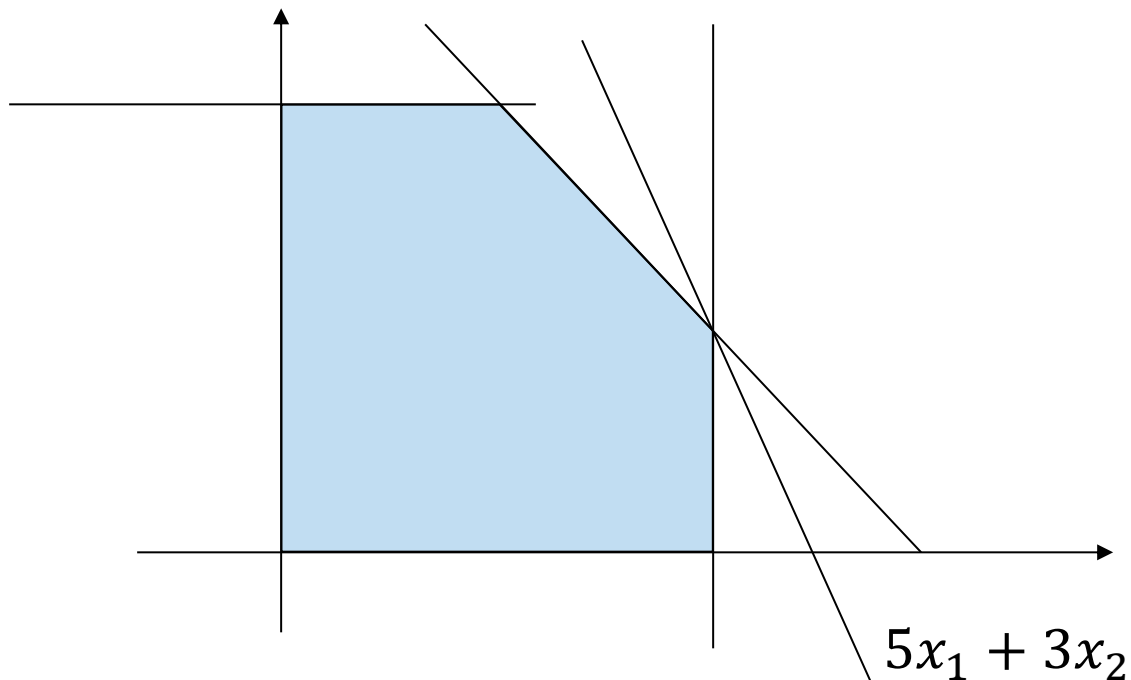
$$N' = \begin{pmatrix} 1 & 0 \\ -3/2 & 1/2 \\ 3 & -1 \end{pmatrix}$$

$$c'_N = -\begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & -3/2 & 3 \\ 0 & 1/2 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} -3/2 \\ 3/2 \end{pmatrix} \begin{matrix} x_3 \\ x_5 \end{matrix}$$

→ The solution is no longer optimal.

Example

What values can the profit of the first product assume such that the current production level remains optimal?



Sensitivity analysis for the solution optimality

Suppose we change the cost of the first product from 5 € to 5 € + δ €, i.e. c changes.

What conditions must δ satisfy?

$$-(c_N - N'^T c_B) \geq 0 \quad (B^{-1}b \text{ is unchanged})$$

$$c'_N = - \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & -3/2 & 3 \\ 0 & 1/2 & -1 \end{pmatrix} \begin{pmatrix} 5 + \delta \\ 3 \\ 0 \end{pmatrix} \geq 0$$

$$\Leftrightarrow 5 + \delta - 4.5 \geq 0$$

$$\Leftrightarrow \delta \geq -0.5$$

Reduced costs

Positive coefficients corresponding to the structural variables are also called **reduced costs**.

- Amount by which the coefficient of the nonbasic variable in the objective function must be reduced (minimization) / increased (maximization) so that the structural variable (e.g., product) is included in the optimal solution.

Basic variables	Coefficients							Solution
	z	x_1	x_2	x_3	s_4	s_5	s_6	
z	1		5			10	10	280
s_4			-2		1	2	-8	24
x_3			-2	1		2	-4	8
x_1		1	1			-1	2	2

Reduced costs in Gurobi

Query the reduced costs:

```
x1.get(GRB.DoubleAttr.RC)
```

Query the smallest objective function value for which the basis remains optimal:

```
x1.get(GRB.DoubleAttr.SAObjLow)
```

Example for reoptimization

$$\begin{array}{llllll} \max & 5x_1 & + & 3x_2 & & \\ \text{s.t.} & x_1 & & & + & x_3 & = & 4 \\ & & 2x_2 & & + & x_4 & = & 12 \\ & 3x_1 & + & 2x_2 & & & + & x_5 & = & 18 \\ & & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0 \end{array}$$

The solution to this LP is $(4, 3, 0, 6, 0)$ with basic variables $\{x_1, x_2, x_4\}$ (i.e. products 1 and 2 are produced) and objective function value of 29.

The company is now considering producing a third product:

A window with

- a price of 4 €
- 4 hours working time on machine 2
- 3 hours working time on machine 3

Reoptimization

By introducing a new variable x_6 we can formulate a new LP with three products:

$$\begin{array}{llllllll}
 \max & 5x_1 & + & 3x_2 & & & + & 4x_6 \\
 \text{s.t.} & x_1 & & & + & x_3 & & = 4 \\
 & & & 2x_2 & & + & x_4 & + 4x_6 = 12 \\
 & 3x_1 & + & 2x_2 & & & + & x_5 + 3x_6 = 18 \\
 & & & & & & & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0
 \end{array}$$

Do we need to solve the LP from scratch?

Possibility: Use the current optimal basis for the new problem.

Reusing the old tableau

We know the optimal tableau for the production problem with two products:

Basic variables	Coefficients							Solution
	z	x_1	x_2	x_3	x_4	x_5	x_6	
z	1			0.5		1.5	?	29
x_1		1		1		0		4
x_4				3	1	-1		6
x_2			1	-1.5		0.5		3

What are x_B, B, B^{-1} and N ?

Computing the reduced costs

We should just check if the variable x_6 is an entering basic variable:

- We compute $c'_N = -(c_N - N'^T c_B)$, where B includes the current basic variables (the current optimal solution) x_1, x_2, x_4 and $N' = B^{-1}N$.

$$x_B = (x_1 \ x_2 \ x_4)^T$$

$$c_B = (5 \ 3 \ 0)^T$$

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{pmatrix}$$

$$x_N = (x_3 \ x_5 \ x_6)^T$$

$$c_N = (0 \ 0 \ 4)$$

$$N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 1 & 3 \end{pmatrix}$$

Computing the reduced costs

$$c_B^T B^{-1} N - c_N^T =$$

$$(5 \quad 3 \quad 0) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 3 & 2 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 1 & 3 \end{pmatrix} - (0 \quad 0 \quad 4)$$

$$(5 \quad 3 \quad 0) \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 3 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 1 & 3 \end{pmatrix} - (0 \quad 0 \quad 4) =$$

$$\left(\frac{1}{2} \quad 0 \quad \frac{3}{2}\right) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 1 & 3 \end{pmatrix} - (0 \quad 0 \quad 4) = \left(\frac{1}{2} \quad \frac{3}{2} \quad \frac{1}{2}\right)$$

Solution

The tableau leads to:

$$c'_N = (0.5 \ 1.5 \ 0.5)$$

$$z + 0.5x_3 + 1.5x_5 + 0.5x_6 = 29$$

$$(\Rightarrow c_B^T B^{-1} b = [0.5 \ 0 \ 1.5] \cdot [4 \ 12 \ 18]^T = 29)$$

x_3, x_5 and x_6 remain non-basic variables.

The current basis is optimal, even after introducing the variable for the new product.

→ The new product is not profitable enough to remove resources for product 1 and 2 and allocate them to the new product.

Reoptimization

In general, the introduction of a new variable does not require a completely new optimization.

One can take the optimal basis of the first solution as the initial basis for the new formulation and run the Simplex algorithm.

The introduction of new constraints is more difficult, since they can cause infeasibilities. However, using the dual (see next class) allows you to treat the new constraint as a new column in the dual program.

Computing profitable products

We can immediately determine the reduced cost of the product at the current basis based on the characteristics of the product (contribution margin + machine hours requirement).

For new product i : $c_N - (B^{-1}N)^T c_B$

$$c_i - \left(\begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ 3 & 1 & -1 \end{pmatrix} \begin{pmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{pmatrix} \right)^T \begin{pmatrix} 5 \\ 3 \\ 0 \end{pmatrix}$$

Only products, for which this value is positive, increase the profit and are included in the production plan.



Summary

(Revised) simplex can help us answer many questions:

- What is the impact of changes on the right side?
- How much money should we invest in expanding our resources?
- What is the best use of our resources?
- How sensitive is our solution to changes in the objective function?
- Is it worthwhile to produce a new product?